Tibor Pál

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Research Interests

My research develops and applies score-driven time-varying parameter and state-space models in macroeconomics and finance, and semi-parametric volatility models using high-frequency data. I analyze and estimate latent macroeconomic variables, including the natural rate of interest and measures of uncertainty, as well as macro-financial interactions, speculative dynamics in asset markets, monetary policy transmission, and inflation distribution dynamics. I also extend structural VAR and dynamic multiple-quantile frameworks to capture nonlinear and state-dependent relationships in advanced economies.

Education

2022–2025	Ph.D. in Statistical Methods – Economics and Policy Analysis of Markets and Firms, University of Salerno (*expected completion: Spring 2026*). *Focus: Score-Driven Models, State-Space Models, Volatility Modeling, Dynamic multiple quantile models*
2024–2025	Visiting Ph.D. Student, University of Malaya. *Focus: Semi-Parametric Volatility Modeling.*
2019–2022	Ph.D. in Economics (*incomplete; transferred to current program*), University of Valladolid. *Focus: Macroeconometrics, Monetary GE Models.*
2021–2022	Visiting Ph.D. Student, University of Verona. *Completed first-year Ph.D. coursework.*
2011–2015	M.A. in International Business (English), Krakow University of Economics. *Thesis: The Evolvement of the Spanish Housing Boom in the Context of EMU Membership.*
2005–2010	B.A. in Finance and Economics, Károly Róbert College. *Thesis: The U.S. Mortgage Crisis and Its Aftermath.*
2009	Academic Scholarship (English), University of Economics, Bydgoszcz, Poland.

Research Projects

1	**Estimating the R-Star in the US: A Score-Driven State-Space Model with Time-Varying Volatility Persistence**. with Giuseppe Storti (University of Salerno). *Working paper.* *Keywords:* R-Star, State-Space Models, Kalman filter, Score-Driven Models.
2	**The Realized Dynamic Score Exponential GARCH**. with Giuseppe Storti (University of Salerno) and Ng Kok Haur (University of Malaya). *Keywords:* realized EGARCH, Score-driven model, Component models, Value-at-Risk, Expected Shortfall.
3	**Multiple Quantile Dynamics of the US Phillips Curve with Time-Varying Parameters**. *Keywords:* Dynamic multiple quantile model, Phillips curve, Inflation Dynamics.

4	**Nonlinear Phillips Curve Dynamics in Japan: Evidence from a Multiple Quantile Approach**. *Keywords:* Dynamic multiple quantile model, Phillips curve, Inflation Dynamics.

Publications and Working Papers

2025	Pál, Tibor and Storti, Giuseppe. *Estimating the R-Star in the US: A Score-Driven State-Space Model with Time-Varying Volatility Persistence.* Working paper, July.
2023	Gutiérrez-Diez, Pedro J. and Pál, Tibor. *Monetary Policy Models: Lessons from the Eurozone Crisis.* Humanities and Social Sciences Communications (Nature), October.
2018	Pál, Tibor. *The Effects of Monetary Policy on House Prices in Spain: The Role of Economic and Monetary Union Membership in the Housing Bubble.* Central European Review of Economics and Management, June.

Teaching Experience

2023	Econometrics (MSc) – University of Verona (English)
2022	Macroeconomics (BSc) – AGH University of Krakow (English)

Presentations

2025	*Estimating the R-Star in the US: A Score-Driven State-Space Model with Time-Varying Volatility Persistence.* December: World Finance Banking Symposium, Brno. November: Warsaw Economic Seminars, Warsaw. July: Int. Society for the Advancement of Financial Economics (ISAFE), Beijing. May: Seminar at the University of Malaya, Kuala Lumpur.
2024	*Estimating the Natural Rate of Interest in the US: An Accelerating Score-Driven State-Space Model.* August: 7th International Conference on Econometrics and Statistics (EcoSta), Beijing. July: Int. Society for the Advancement of Financial Economics (ISAFE), Pattaya.
2022	*The Real Effect of the (De)Leverage Pressure.* Brownbag Seminar, University of Verona.
2022	*Rethinking Monetary Policy Models: Lessons from the 2008–2018 Crisis in the Eurozone.* Seminar at the University of Windsor (online).
2019	*A Debt Deflation Stylized Model: The Effects of Deleveraging in the Eurozone.* 2nd International Conference on Modern Tendency in Social Science, Singapore.
2018	*The Real Effects of Deleveraging in the Eurozone.* 32nd Conference of International Applied Economics, Huelva.
2018	*The Credit View on the Output Gap and the Concept of the Non-Accelerating Inflation Rate of Credit Expansion (NAIRCE).* 15th International Conference on Developments in Economic Theory and Policy, Bilbao.
2017	*The Effects of Monetary Policy on House Prices in Spain: The Role of the European Economic and Monetary Union Membership in the Latest Housing Bubble.* Fifth International Conference, WSB University, Wroclaw.

Further Trainings and Attendances

2023	Methods in Time Series Econometrics, Bertinoro. *Summer school by Francesco Bianchi and Karel Mertens.*
2023	Financial Time Series and High-Frequency Econometrics, Perugia. *Postgraduate course by Massimiliano Caporin, Walter Distaso, among others.*
2023	Tenth Italian Congress of Econometrics and Empirical Economics, Cagliari.
2023	Financial Time Series and Tail Risk Forecasting with Python, University of Salerno. *Workshop by Richard Gerlach.*
2022	Padova Macro Talks, University of Padova. *Workshop organized by Efrem Castelnuovo.*

Work Experience

2018–2024	Omni Calculator (part-time). *Financial Specialist, Reviewer, and Advisor.*
2014–2017	Shell BSC. *Financial Analyst.*
2010–2014	IBM BTO. *GL Accountant.*

Skills

Technical: MATLAB, R, Stata, Python, JavaScript.

Languages: English (C2), Spanish (B1), Italian (A1), Polish (A1), Chinese (A1), Hungarian (native).